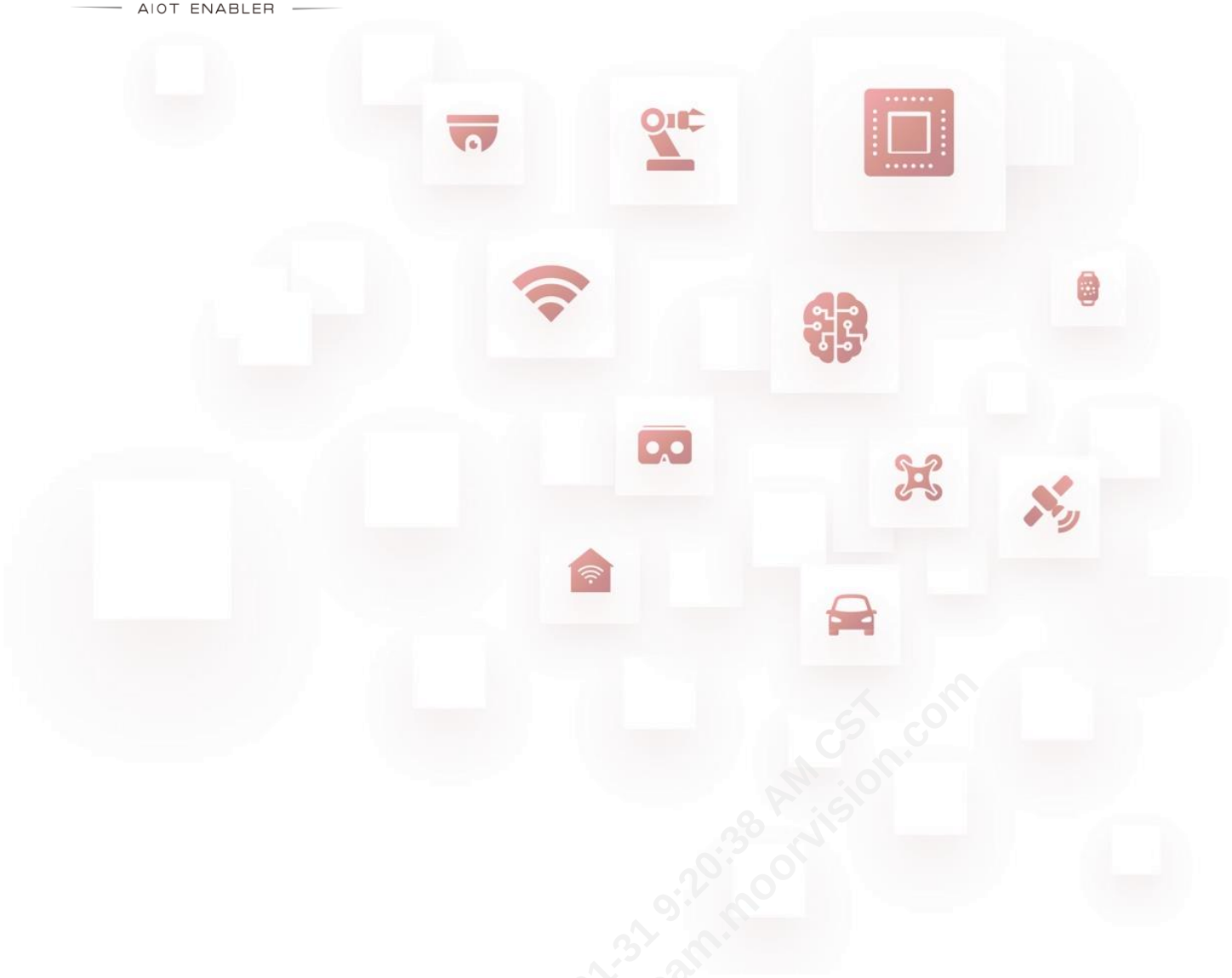
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Stefan_ecam.moorvision.com

Thundercomm TurboX Flash for Windows

User Guide

Rev. V1.2
Dec 10, 2024

DN: tc-A-163612-en

Revision History

Revision	Date	Author	Description
1.0	Sep 15, 2023	Lining Jiao	Initial release.
1.1	Sep 27, 2023	Lining Jiao	• Swap the positions of 3.1. Flash the image and 3.2. UFS Provision. • Update Step 2 of 3.1.5. Flash the FlatBuild Image. • Add introduction at the beginning of 3.2. UFS Provision.
1.2	Dec 10, 2024	Teng Xu	Add a description to the "Erase All Before Download" option in section 3.1.4. Set the download configuration.

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About This Document

- This document describes the usage of Tflash for Windows and is applicable to all modules.
- Illustrations in this documentation might look different from your product.
- Depending on the model, some optional accessories, features, and software programs might not be available on your device.
- Depending on the version of operating systems and programs, some user interface instructions might not be applicable to your device.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of the products, including this guidebook.
- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, `cp armcc armcpp`.
- Code variables appear in angle brackets, for example, `<number>`.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font; on the host computer use $ as shell prompt, while on the target device use # as shell prompt, for example,

```
$ adb devices
# logcat
```

- Part of the code that does not contain instructions appear in a different format, for example,

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
```

- Folders, path and files are formatted in italic, for example, *turbox_flash_flat.sh*.

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Chapter 1. Overview

Tflash (TurboX Flash) is an image burning tool developed by Thundercomm. It can be used to flash FlatBuild onto Thundercomm SOM modules. Tflash includes two versions: Windows graphical interface version and LU command-line version. This document will introduce the usage of the Windows graphical interface version.

FlatBuild is an image format released by Thundercomm, which can be flashed in EDL (Emergency Download) mode. When a device is unable to boot due to software issues, you can repair it by flashing FlatBuild.

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Chapter 2. Install Tflash

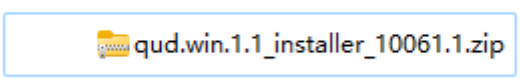
2.1. Install the QUD USB Driver

Running Tflash requires installing the QUD USB driver; otherwise, the device will not be recognized.

2.1.1. Download the QUD USB driver

Download address: https://docs.thundercomm.com/turbox_doc/data/common/tools/qud-win-installer.zip

Downloaded file:



➲ **NOTE:** To download Tflash, you need to log in to https://docs.thundercomm.com. Only Thundercomm staff or users who have purchased Thundercomm products and signed the NDA (Non-Disclosure Agreement) are eligible to access the download.

2.1.2. Install the QUD USB Driver

Step 1. Decompress the downloaded zip file to the target directory.

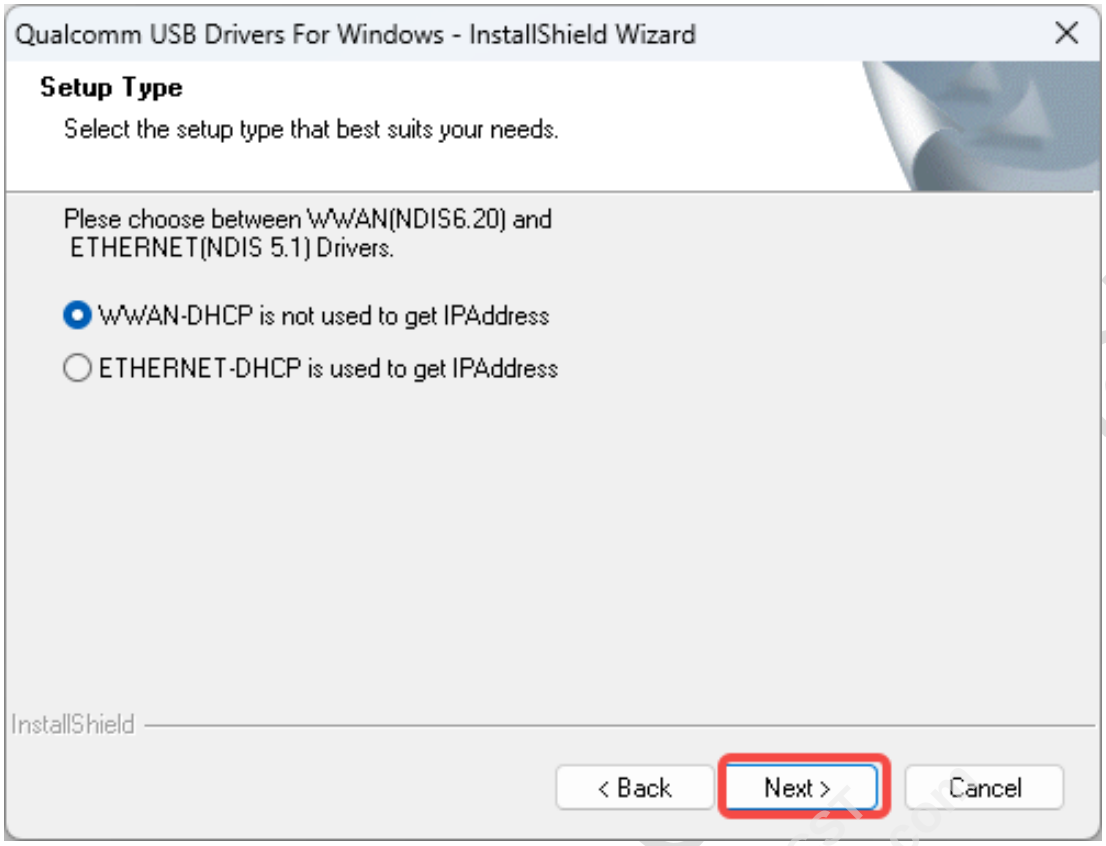
Step 2. Double-click on setup.exe in the extracted directory.

Step 3. Select "Yes" in the User Account Control interface.



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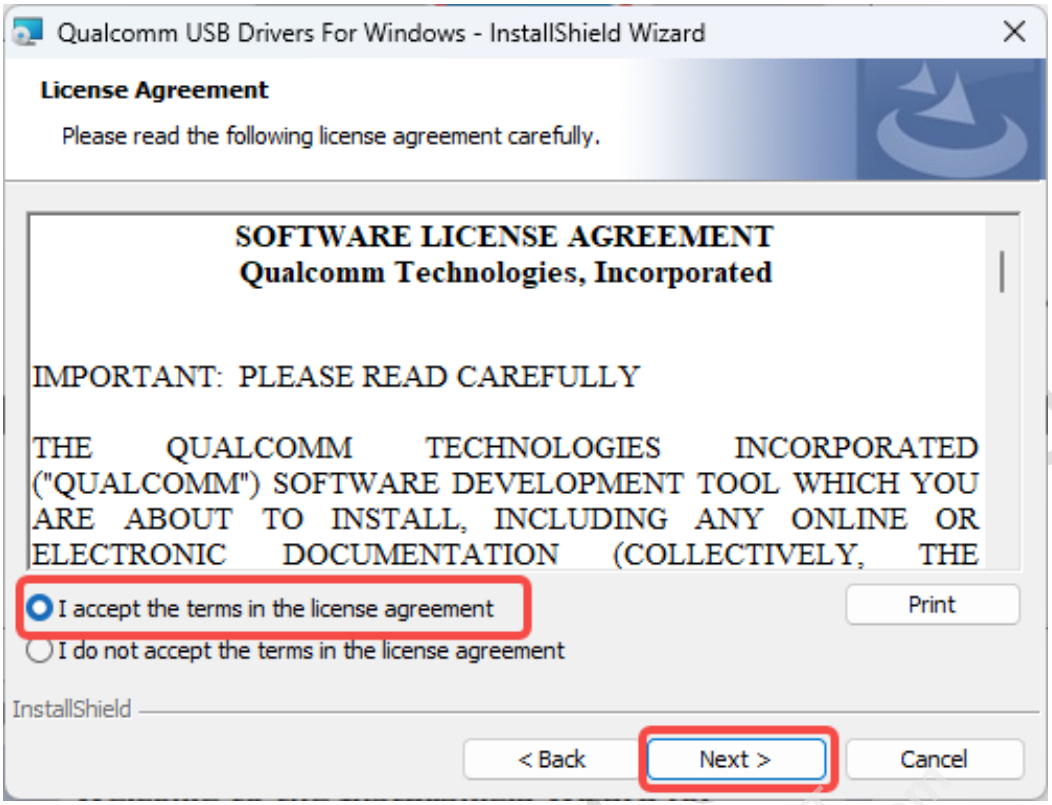
Step 4.Click on "Next".



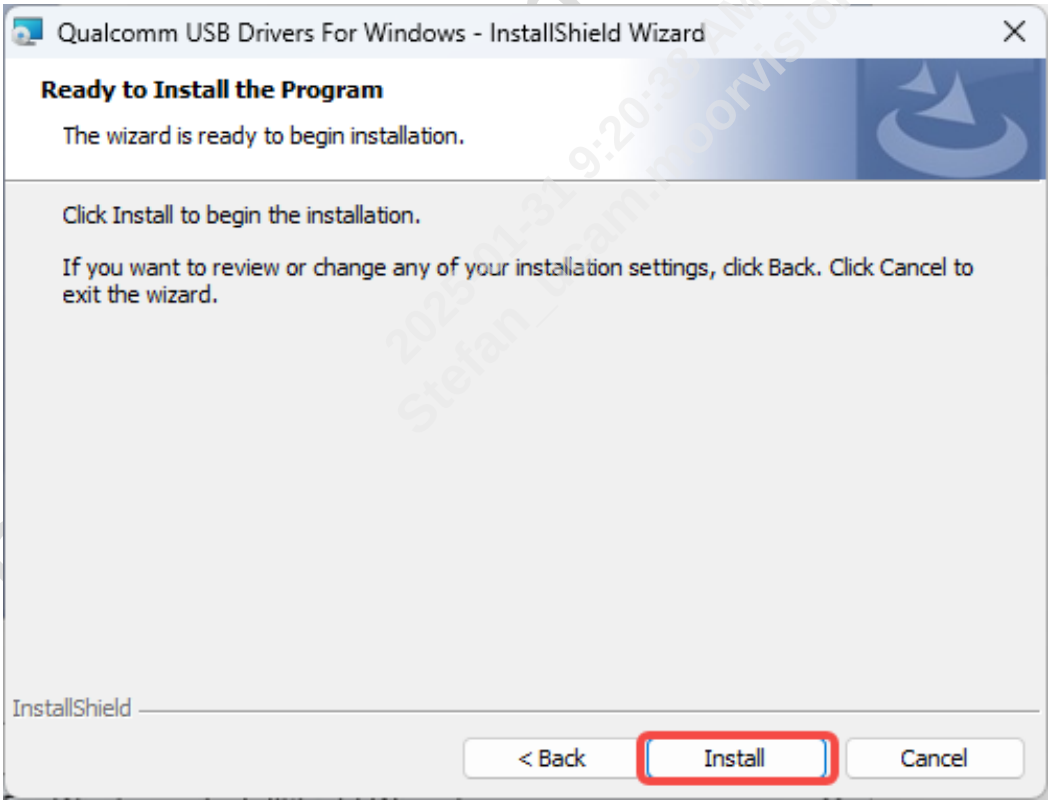
Step 5.Click on "Next" again.



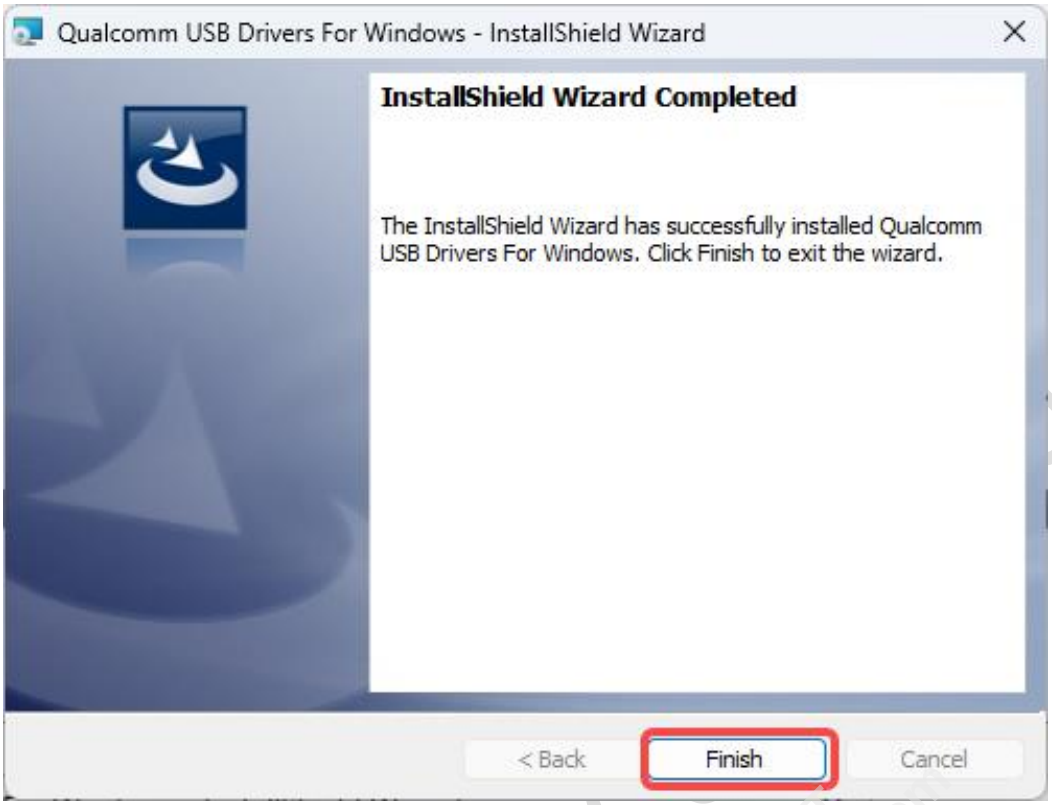
Step 6.Accept the terms in the license agreement and click on "Next".



Step 7.Click on "Install".



Step 8.Click on "Finish".



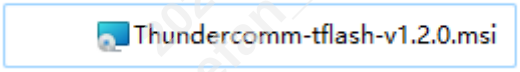
2.2. Install Tflash on PC

2.2.1. Download Tflash installation package

Download address:

https://docs.thundercomm.com/turbox_doc/data/common/tools/thundercomm-tflash-windows.msi

Downloaded file:



➲ **NOTES:**

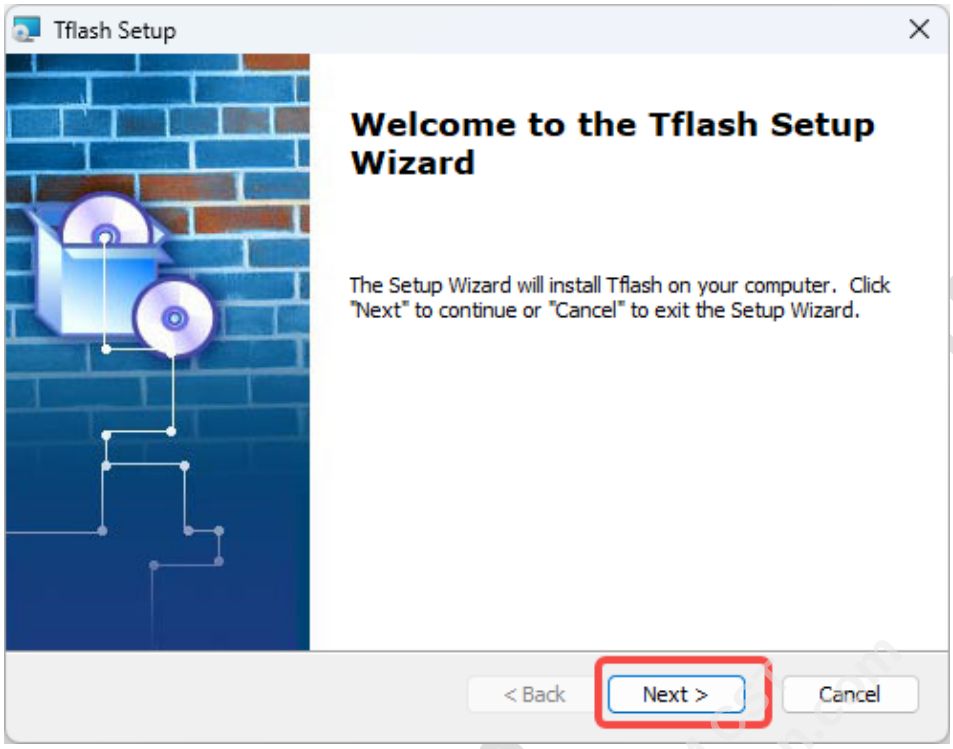
- v1.2.0 stands for the version number, which will change when a new version is released.
- To download Tflash, you need to log in to https://docs.thundercomm.com. Only Thundercomm staff or users who have purchased Thundercomm products and signed the NDA (Non-Disclosure Agreement) are eligible to access the download.

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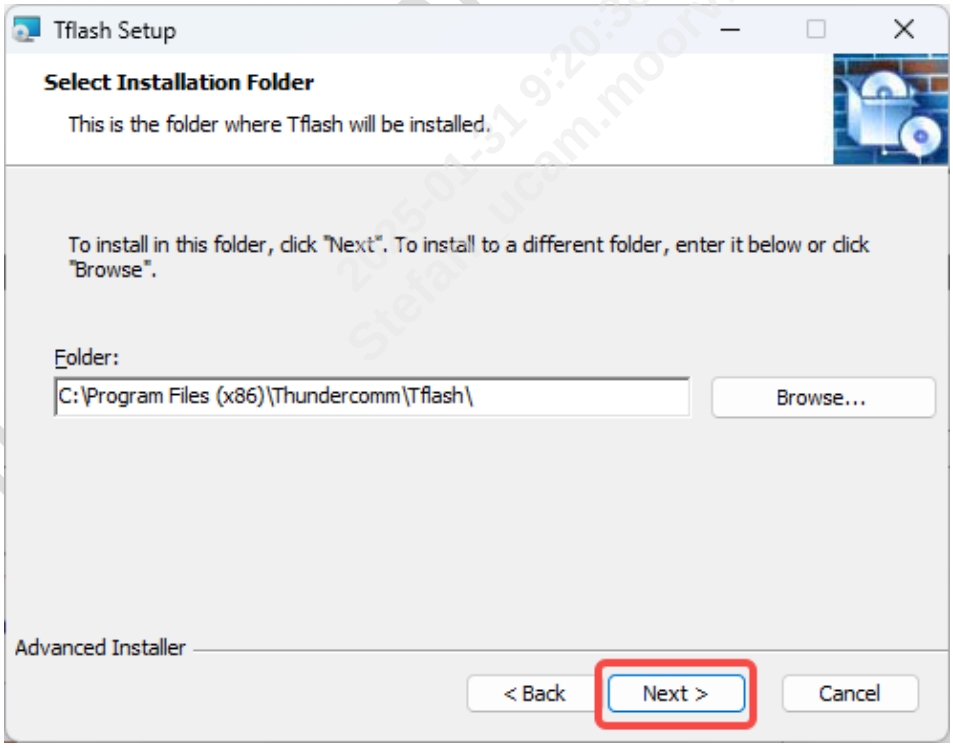
2.2.2. Install Tflash

Step 1. Double-click on Thundercomm-tflash-v1.2.0.msi to start installation.

Step 2. Click on "Next".

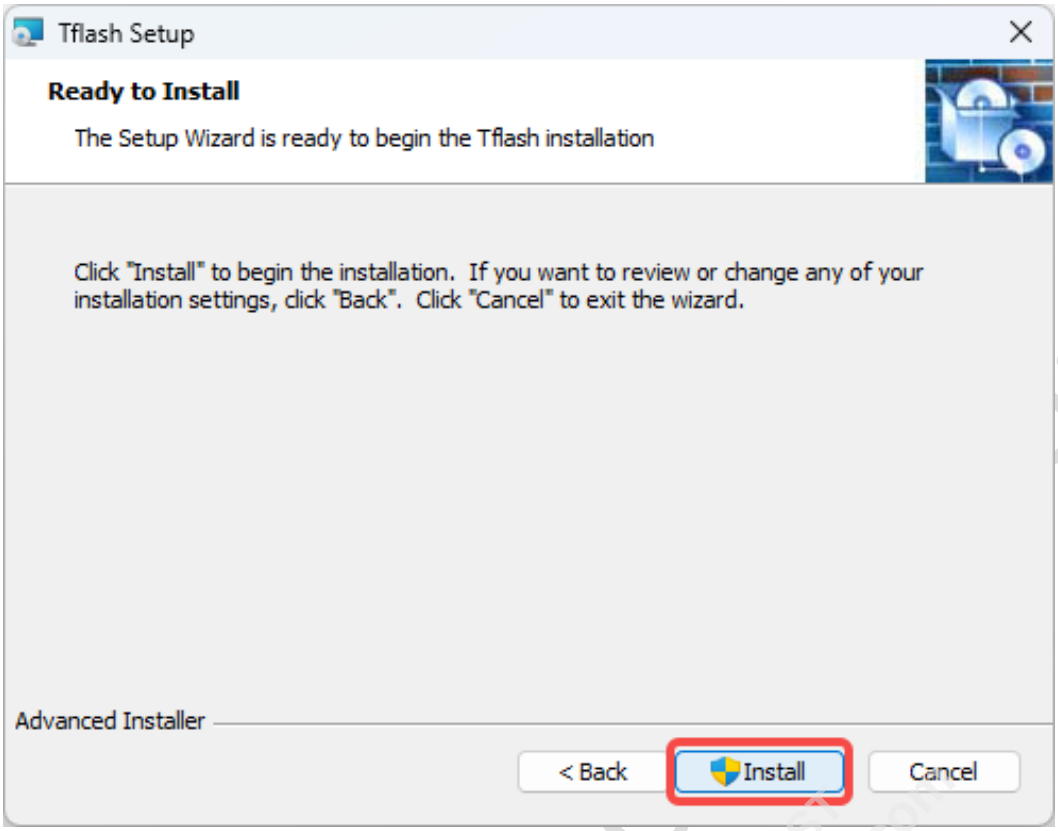


Step 3. Click on "Next" again.

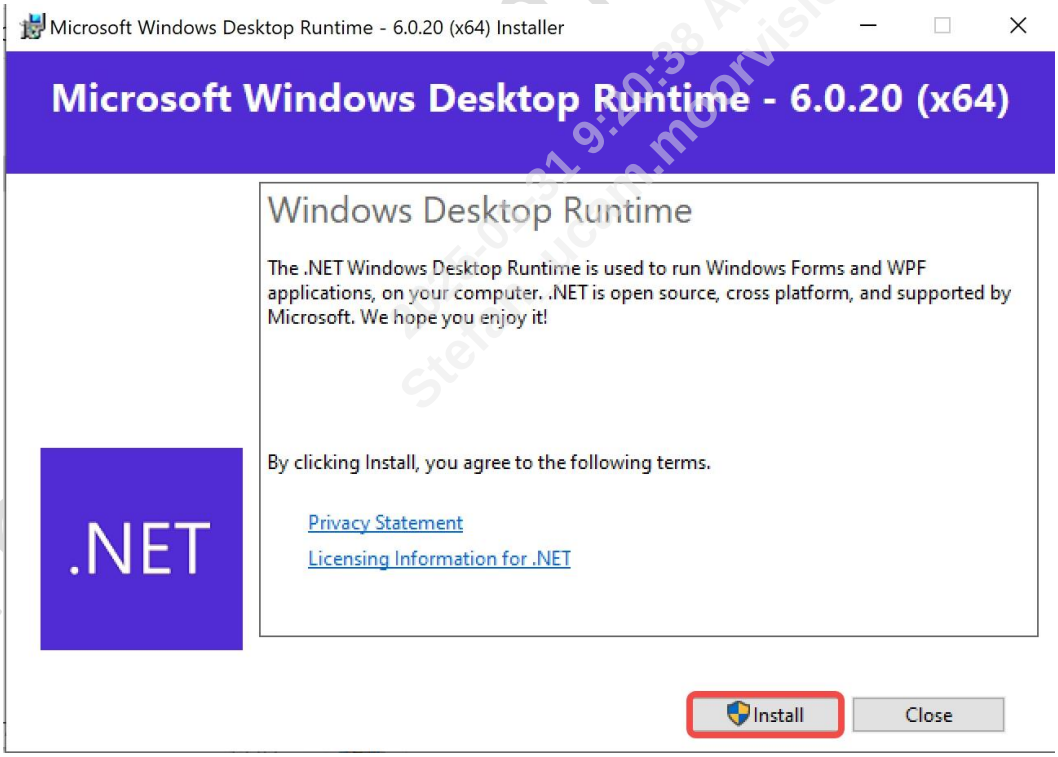


➲ **NOTE:** The installation path can be modified as required.

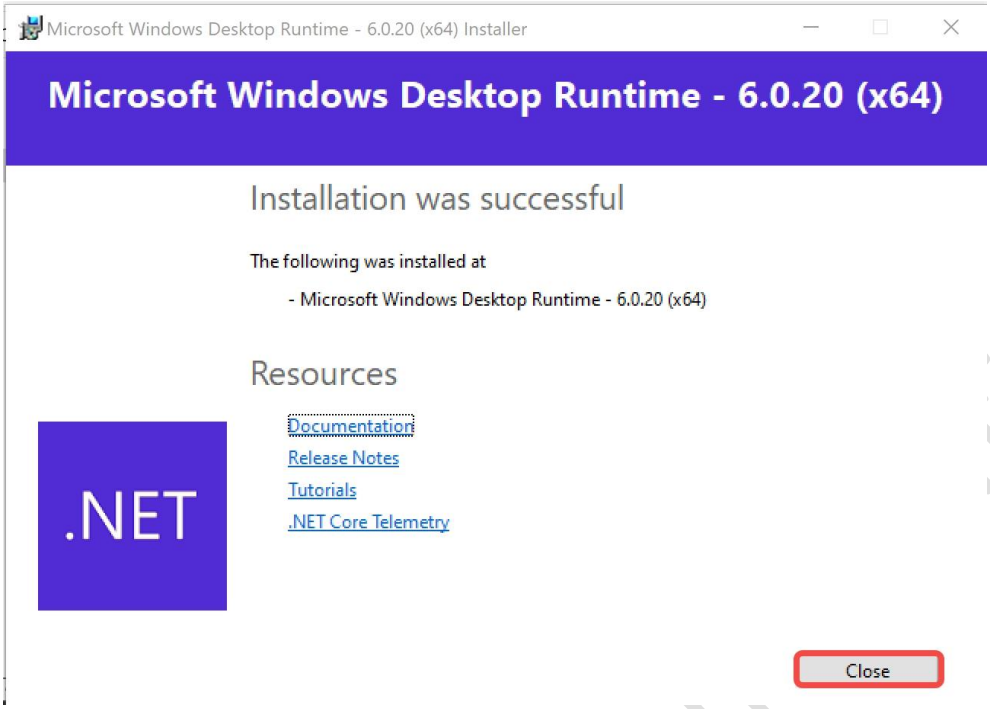
Step 4.Click on "Install".



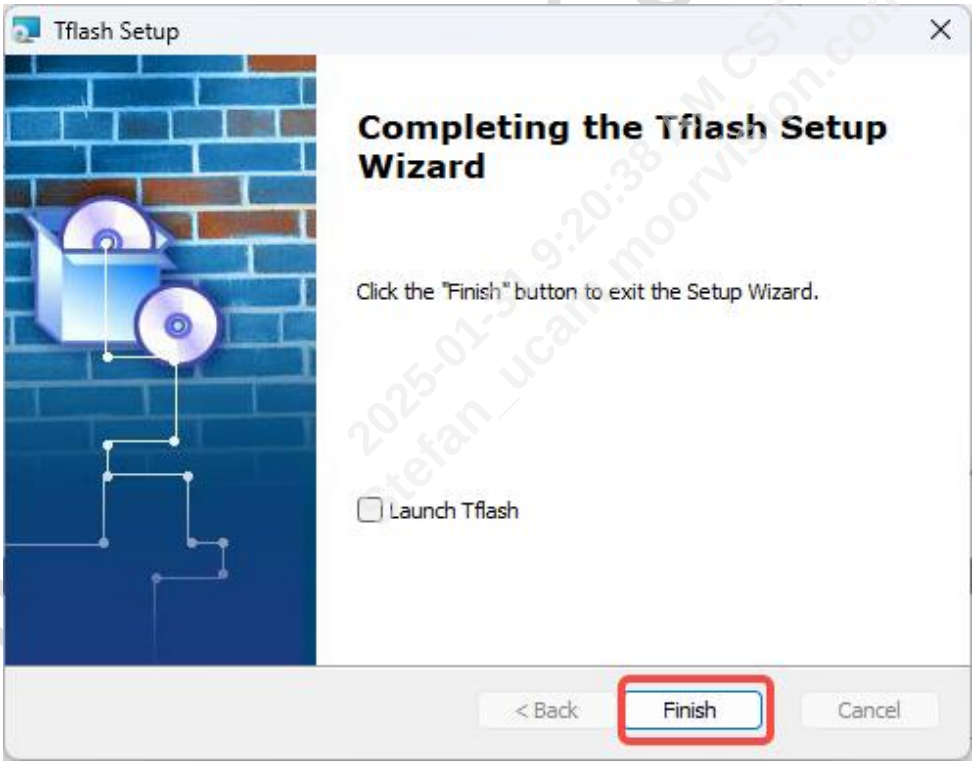
Step 5.Click on "Install" again.



Step 6.Click on "Close".



Step 7.Click on "Finish".



After the installation is complete, the Tflash icon will appear on the desktop and in the "Start" menu:



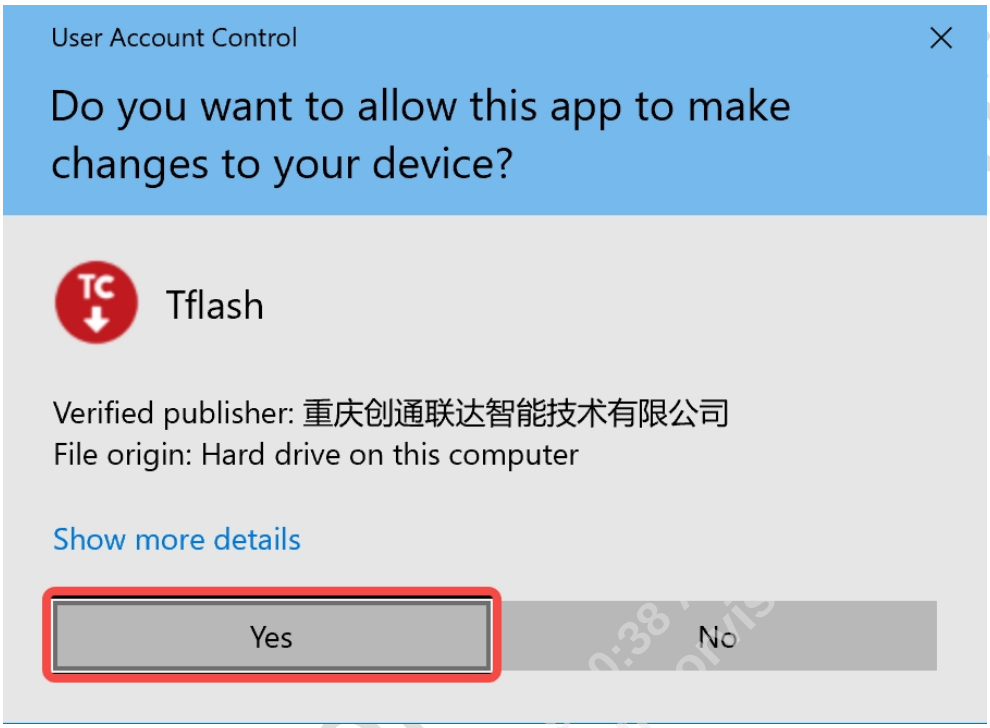
Chapter 3. Flash with Tflash

3.1. Flash the image

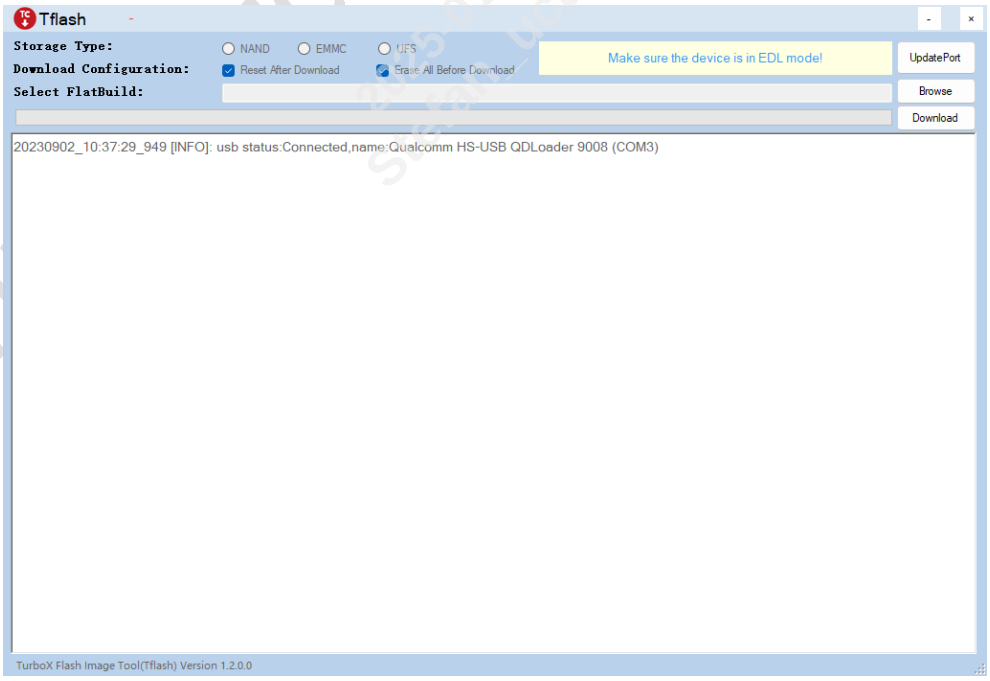
3.1.1. Run Tflash

Step 1.Click on the Tflash icon on the desktop or in the "Start" menu.

Step 2.Select "Yes" in the User Account Control interface that pops up.



Refer to the running interface of Tflash as follows:



3.1.2. Enter 9008 mode

Step 1. Power off the board by disconnecting the power cable and USB cable.

Step 2. Press and hold the **Force_USB_Boot** button (or named "**Force_USB**" on certain development boards).

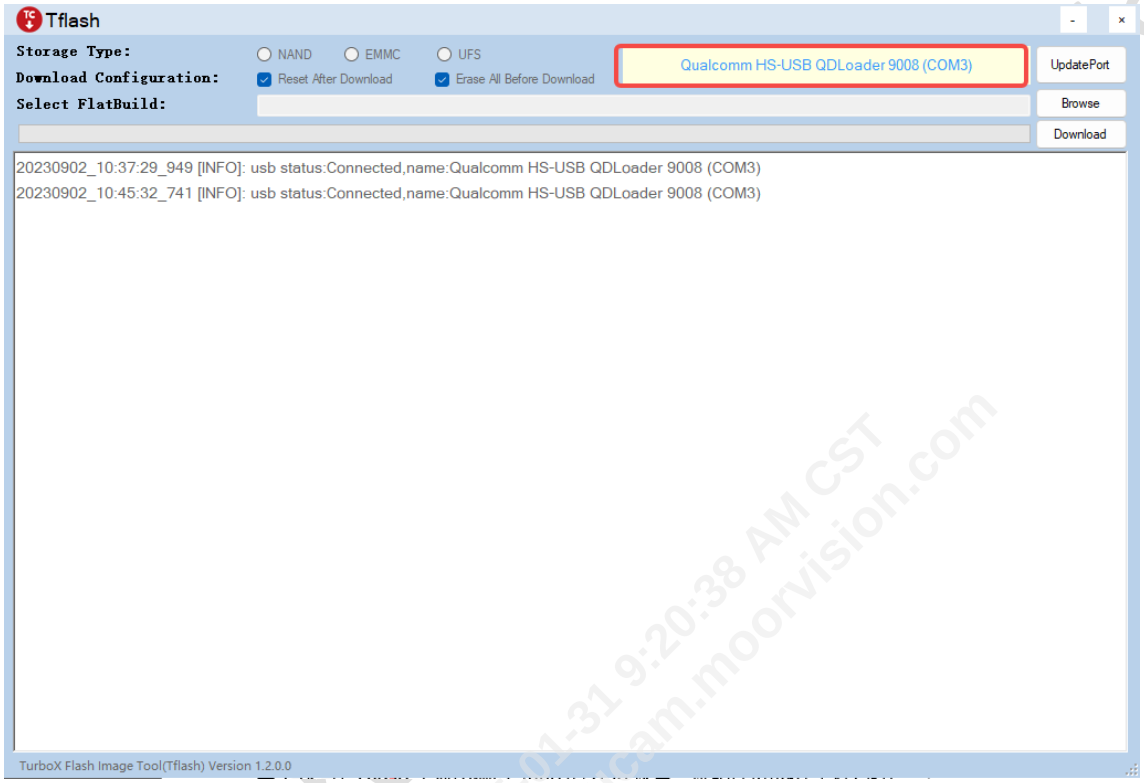
Step 3. Connect the power cable to the board.

Step 4. Long-press (3 seconds) the **Power** button on the board.

Step 5. Connect the board to PC with a USB Type-C cable (or "Micro USB cable" on certain development boards).

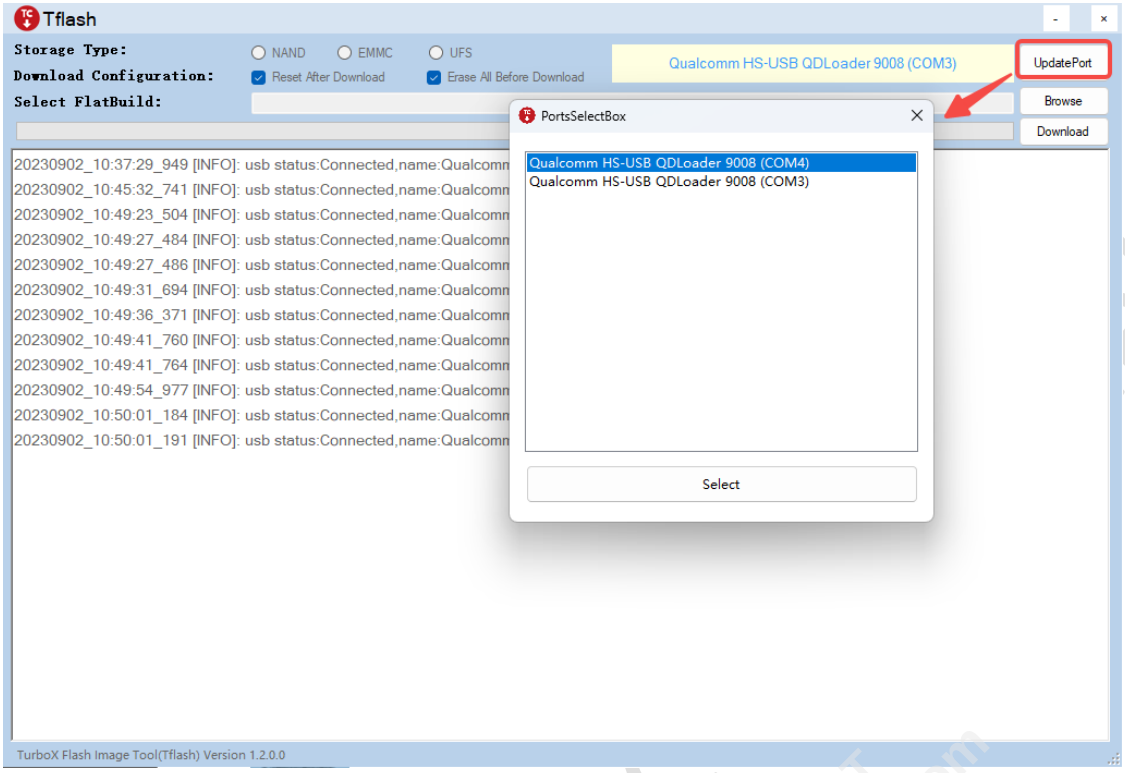
Step 6. Wait for 3 seconds, and then release the **Force_USB_Boot** button.

If Tflash displays a USB device with port **9008**, it indicates a successful connection:



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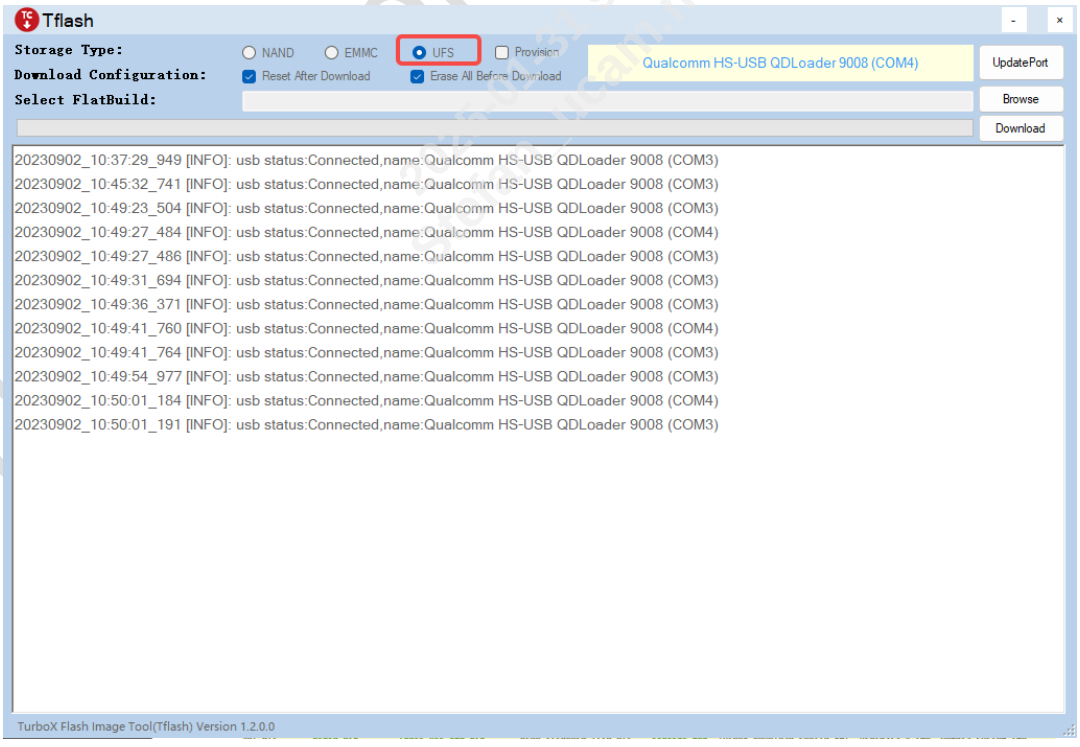
➲ **NOTE:** In **9008** mode, when multiple devices are connected to the PC, you can select the device you want to flash by clicking on "**UpdatePort**":



3.1.3. Select the storage type

Select the correct "Storage Type" based on the storage type of the device you want to flash. Please refer to the relevant SOM datasheet for the storage types of Thundercomm's various SOMs.

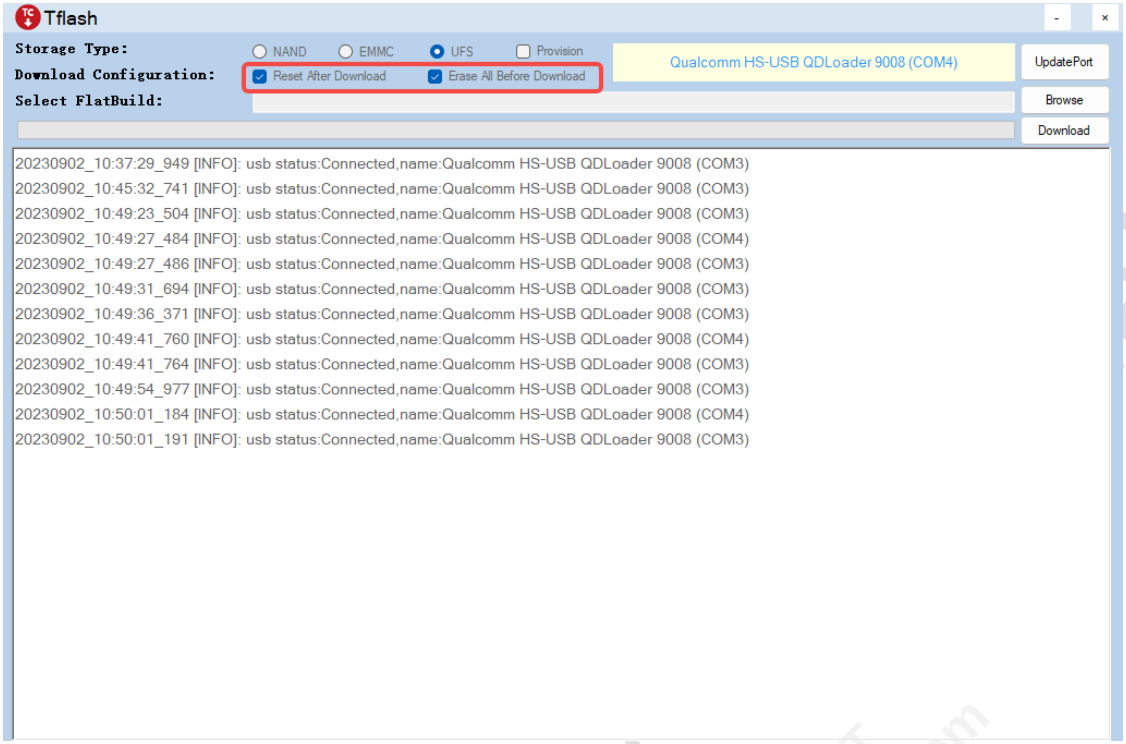
Refer to the following example:



➲ **NOTE:** Select "Provision" when performing UFS provision; deselect it when flashing images.

3.1.4. Set the download configuration

Select "Reset After Download" and "Erase All Before Download" as needed.



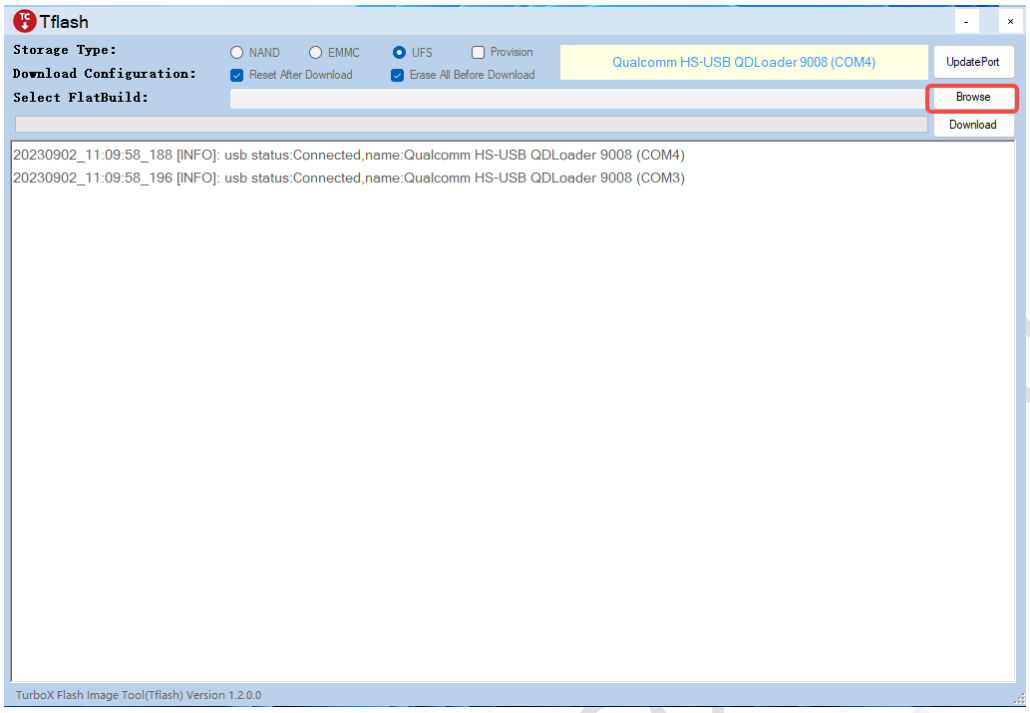
- Reset After Download: With this option checked, the device will automatically restart after the flashing process is completed.
- Erase All Before Download: With this option checked, the data in the device will be erased before flashing. MAC addresses, SN (Serial Number), modem preset parameters, antenna calibration data, and other information in the device will be erased. **If this data has not been backed up beforehand, it cannot be recovered. Do not select this option for products with modem functionality. Otherwise, modem preset parameters and antenna calibration data will be erased, causing the modem fail to operate normally.**

➲ **NOTE:**

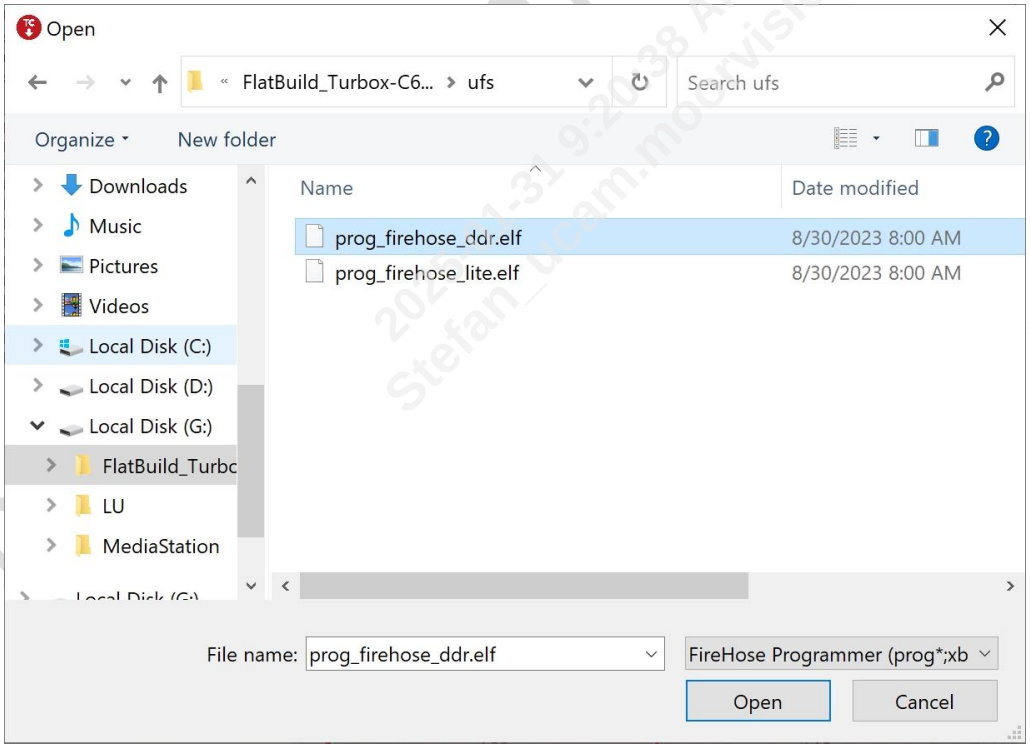
- If the partitions before and after flashing are different, even if “Erase All Before Download” is not checked, MAC addresses and other information will be unusable.

3.1.5. Flash the FlatBuild image

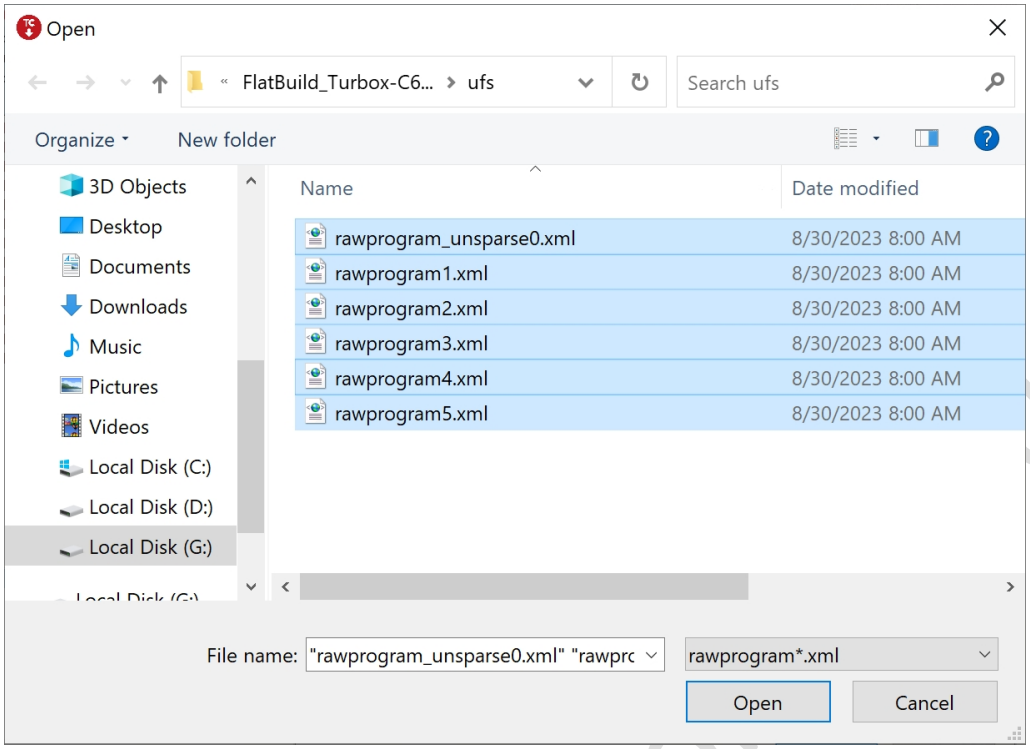
Step 1. Click on "Browse".



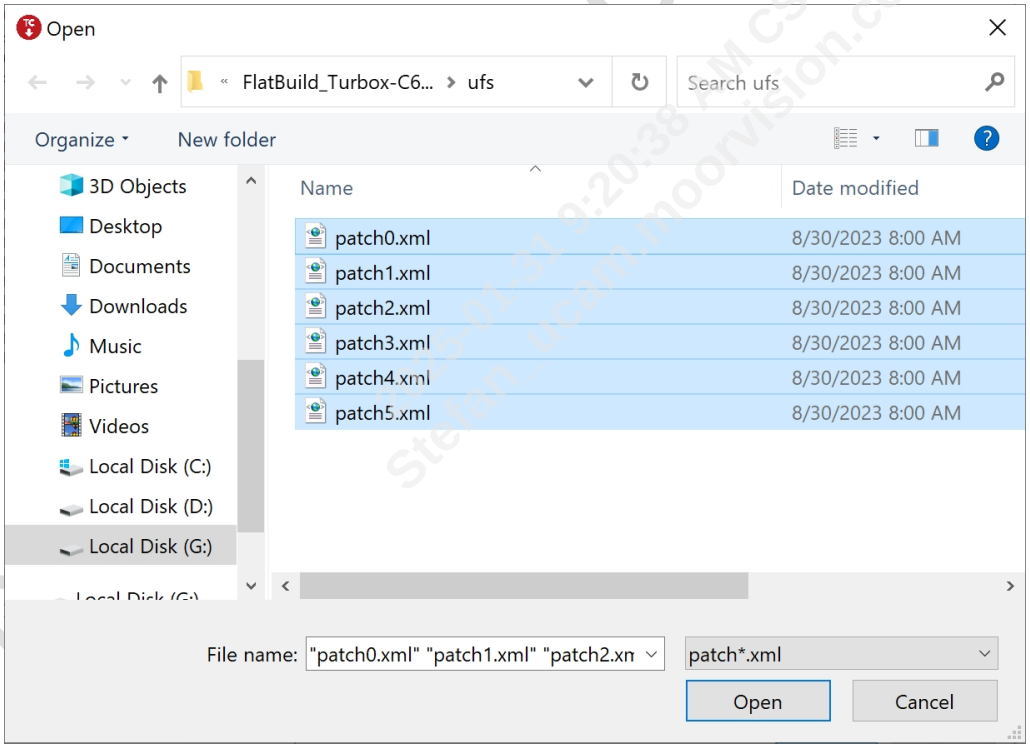
Step 2. Go to the FlatBuild directory, select "Programmer File", and select "prog_firehose_ddr.elf" (or "xbl_s_devprg_ns.melf" on certain platforms; for specific options, please refer to the relevant SDK release note).



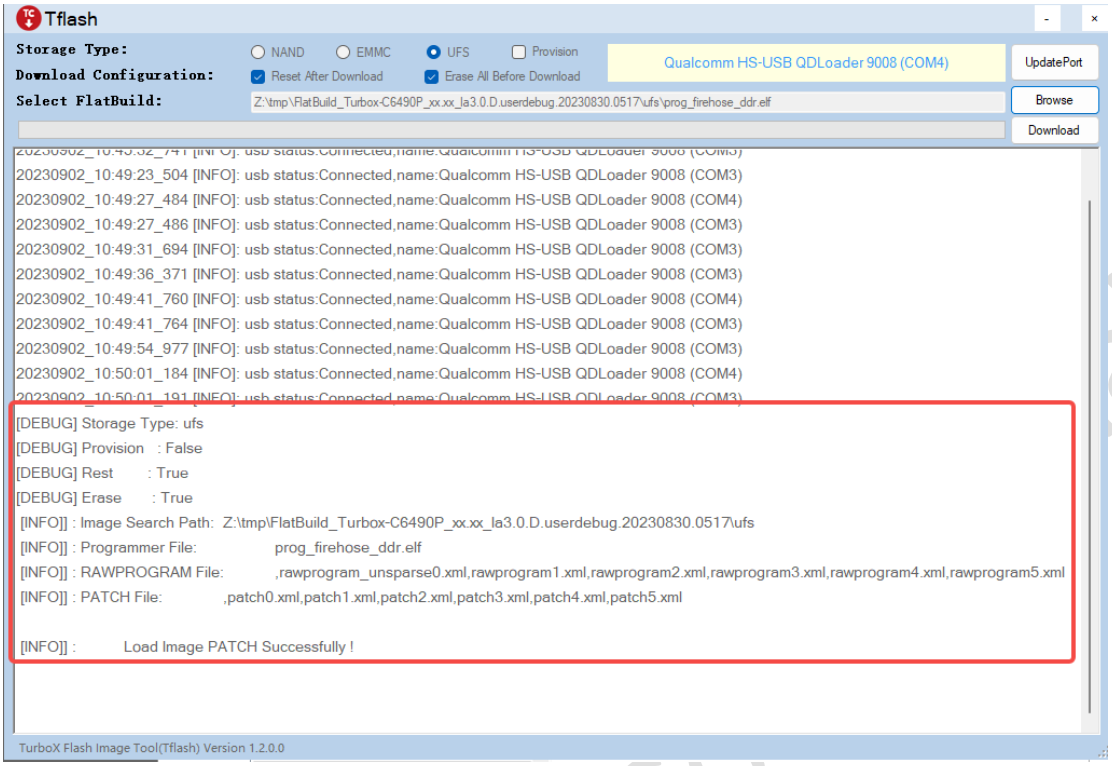
Step 3.Select all the XML files whose names start with "rawprogram".



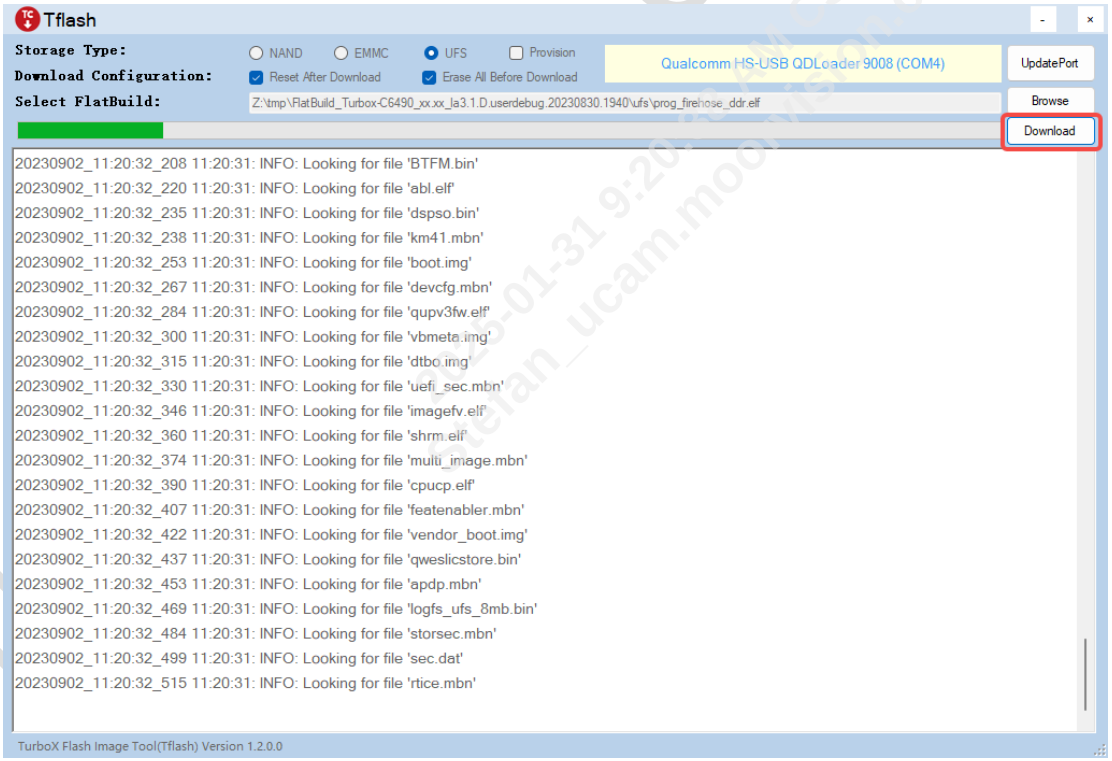
Step 4.Select all the XML files whose names start with "patch".



After the settings are completed, you can obtain the relevant settings from the Tflash interface:

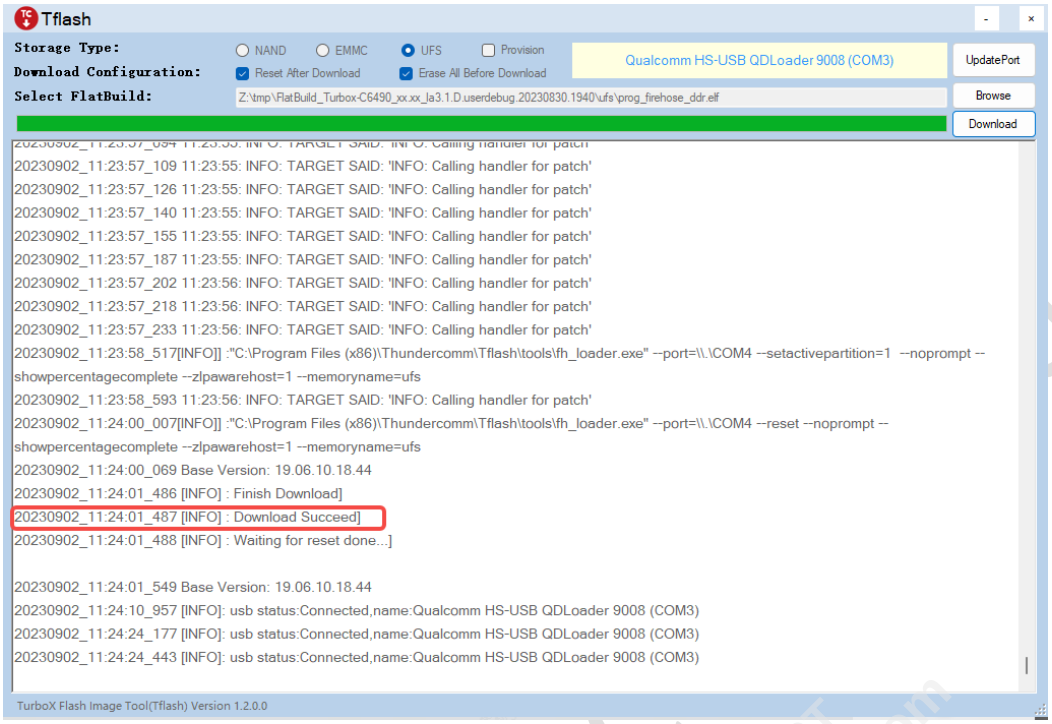


Step 5.Click on "Download" to start flashing.



➲ **NOTE:** The flashing time may vary depending on the device. During the flashing process, you can monitor the progress through the green progress bar.

After successful flashing, the following log will be output:



3.1.6. Restart the development board

- If "Reset After Download" is checked, the development board will automatically restart after the flashing is completed.
- If "Reset After Download" is unchecked, disconnect the power cable and USB cable to power off the development board, then reconnect the power cable and press the **Power** button to start the development board.

➲ **NOTE:** You can use UART log or adb commands to confirm if the development board has started up properly. For detailed instructions, please refer to the user manual of each development board.

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3.2. UFS Provision

UFS Provision divides UFS into several logical units (LUNs). Only after executing UFS Provision, can you proceed with the image flashing.

Thundercomm's SOMs are already provisioned during the manufacturing process, so users generally do not need to perform provisioning again. However, if the software version you need to flash differs from the factory software version of the SOM in terms of LUN requirements, then you need to perform provisioning on the UFS again.

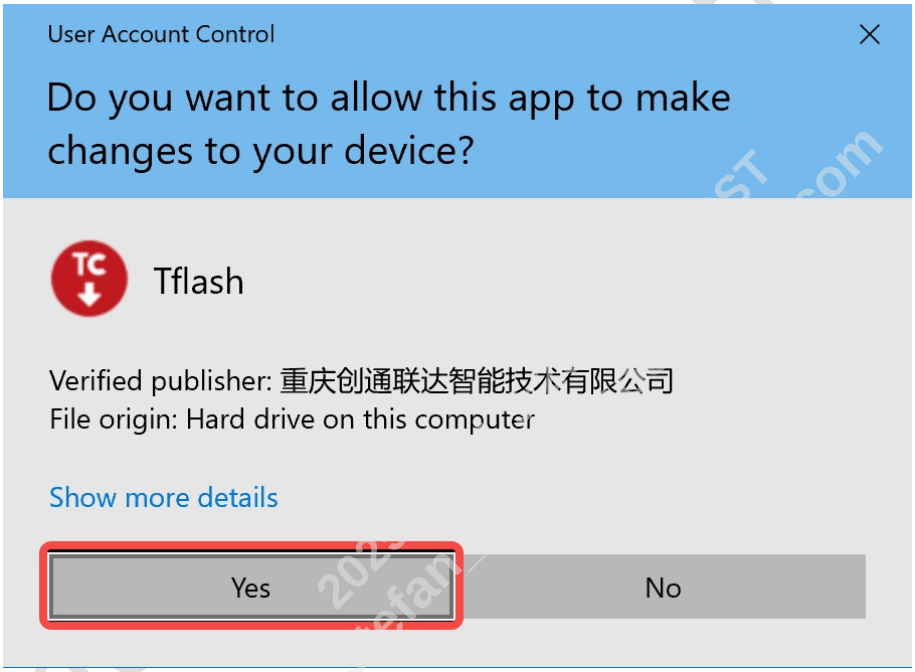
When burning software released by Thundercomm, it is usually unnecessary to perform provisioning. Any cases where provisioning is required will be specifically mentioned in the SDK Release Note.

You can use the Tflash tool to perform provisioning on the UFS. For specific steps, please refer to the following subsections.

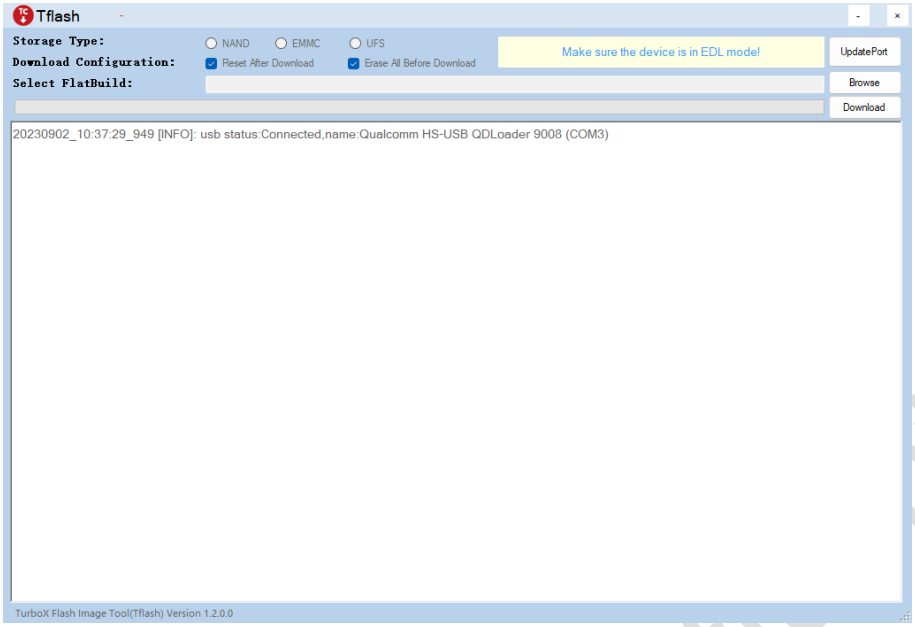
3.2.1. Run Tflash

Step 1. Click on the Tflash icon on the desktop or in the "Start" menu.

Step 2. Select "Yes" in the User Account Control interface that pops up.



Tflash running interface is as follows:



3.2.2. Enter 9008 mode

Step 1. Power off the board by disconnecting the power cable and USB cable.

Step 2. Press and hold the **Force_USB_Boot** button (or named "**Force_USB**" on certain development boards).

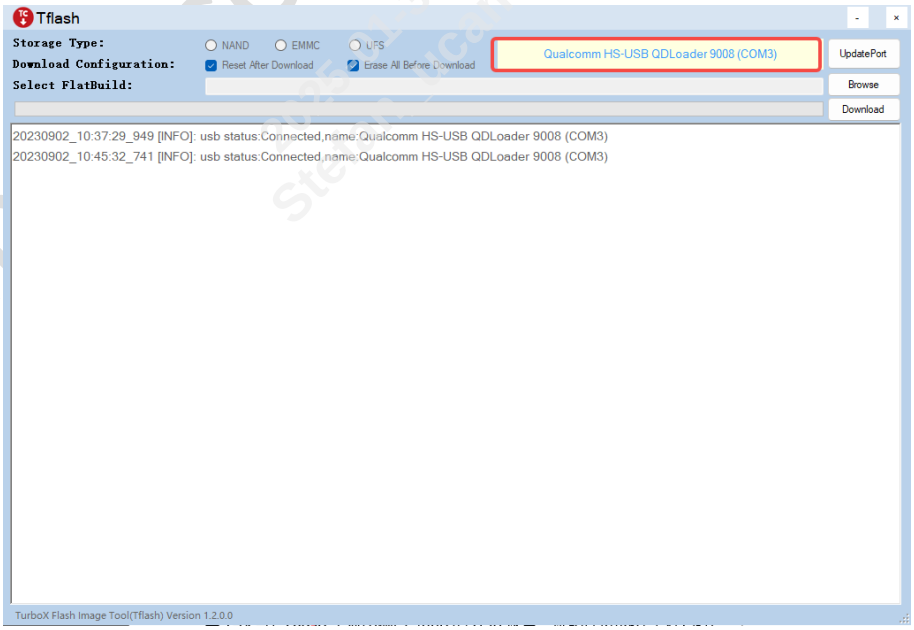
Step 3. Connect the power cable to the board.

Step 4. Long-press (3 seconds) the **Power** button on the board.

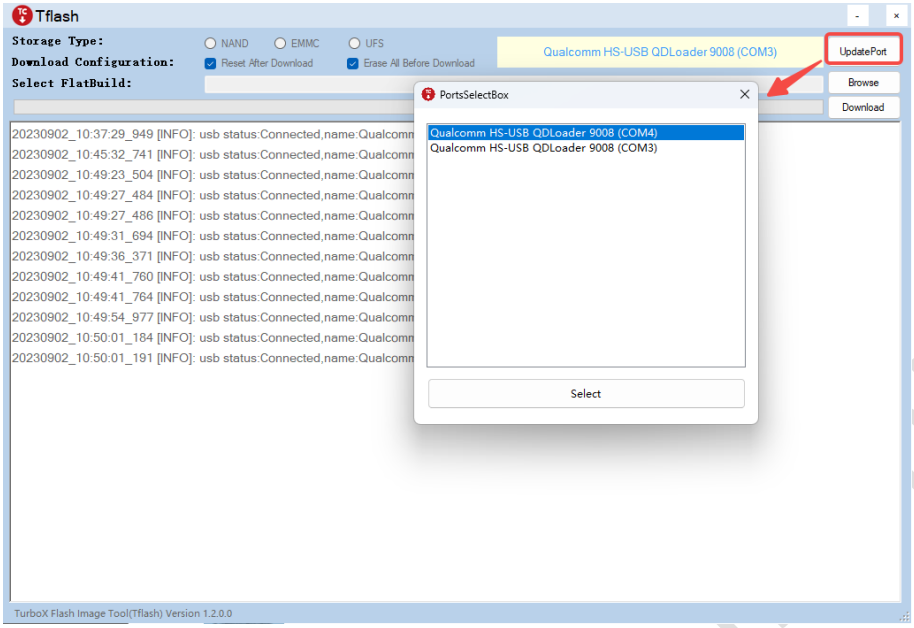
Step 5. Connect the board to PC with a USB Type-C cable (or "Micro USB cable" on certain development boards).

Step 6. Wait for 3 seconds, and then release the **Force_USB_Boot** button.

If Tflash displays a USB device with port **9008**, it indicates a successful connection:

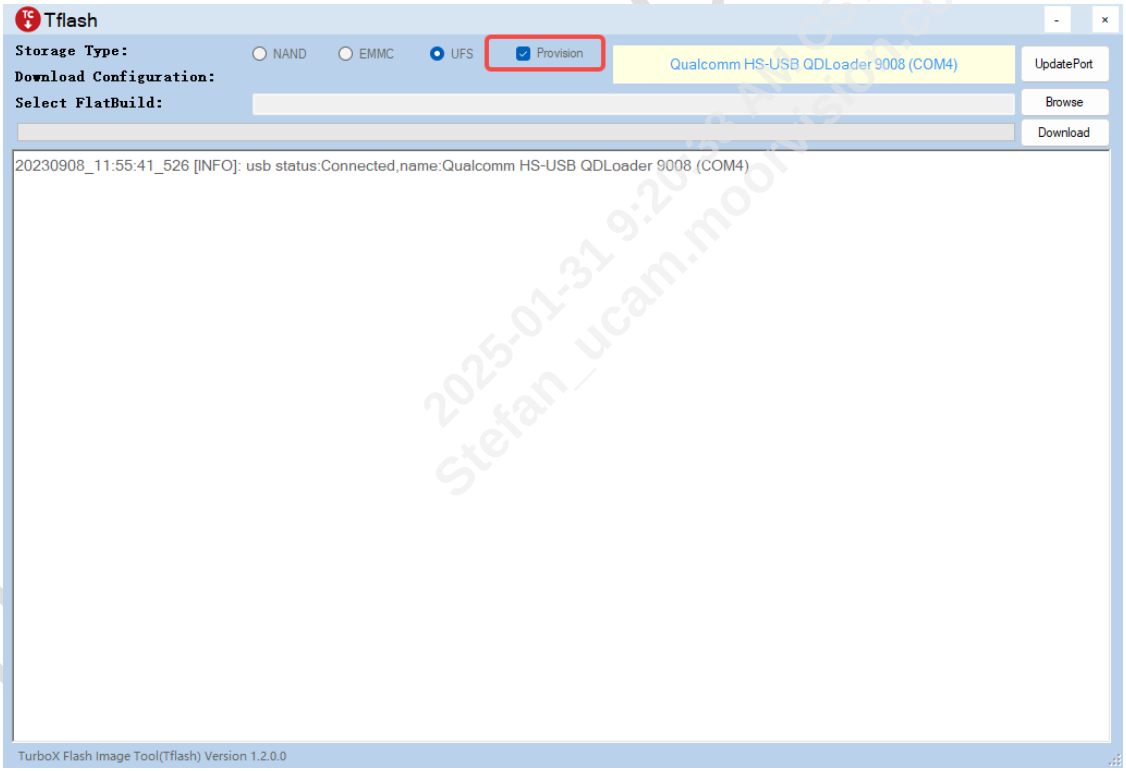


➲ **NOTE:** In **9008** mode, when multiple devices are connected to the PC, you can select the device you want to flash by clicking on "**UpdatePort**":



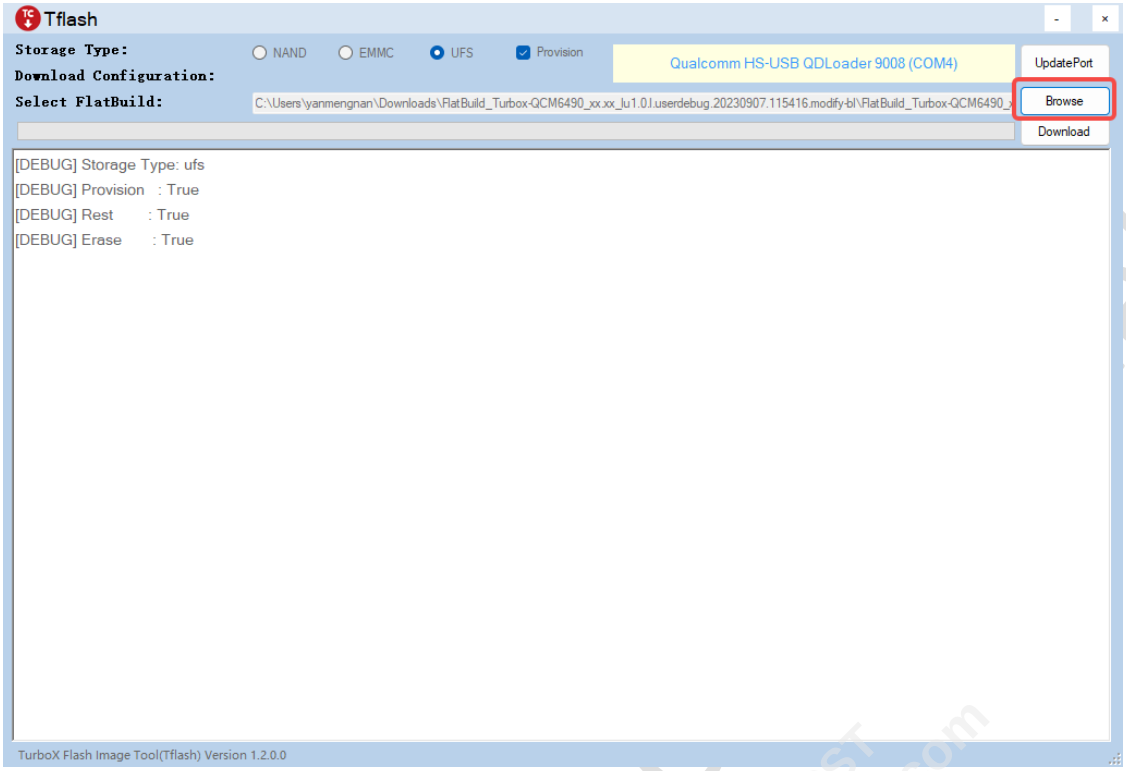
3.2.3. Select the storage type

Select "UFS" in the "Storage Type" and check "Provision".



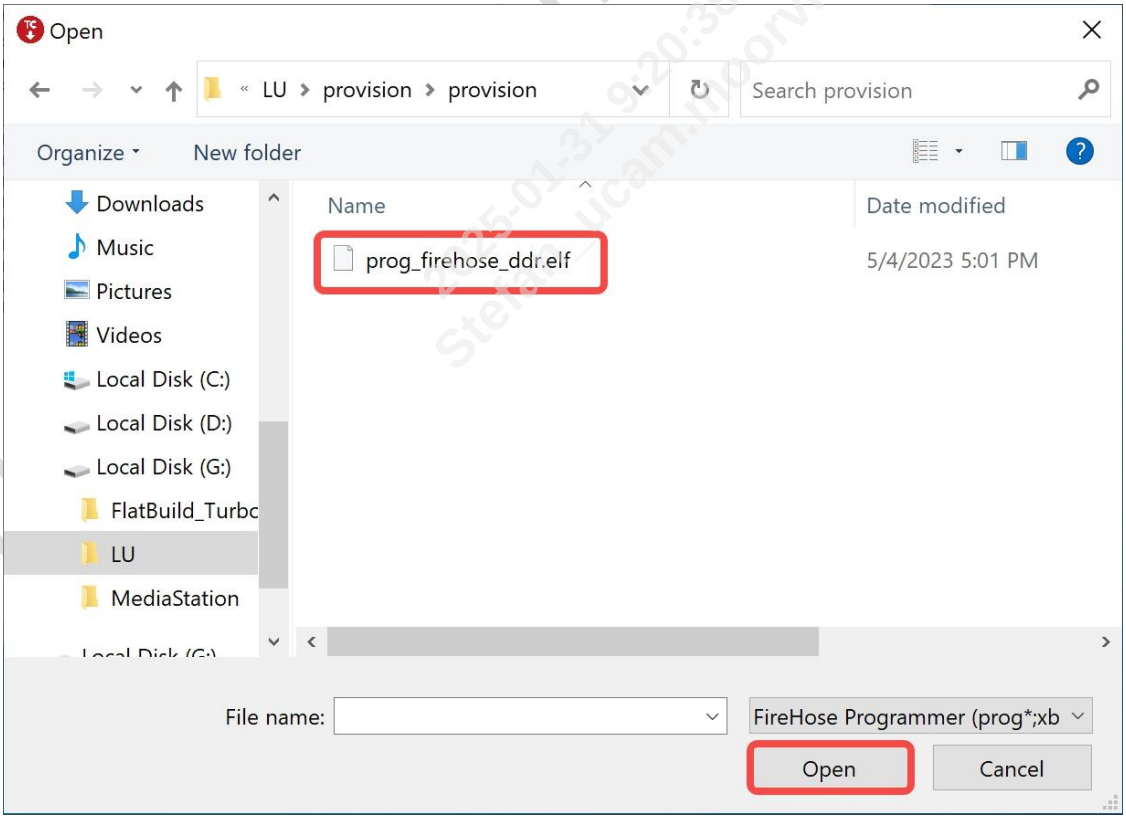
3.2.4. Flash the provision file

Step 1.Click on "Browse".

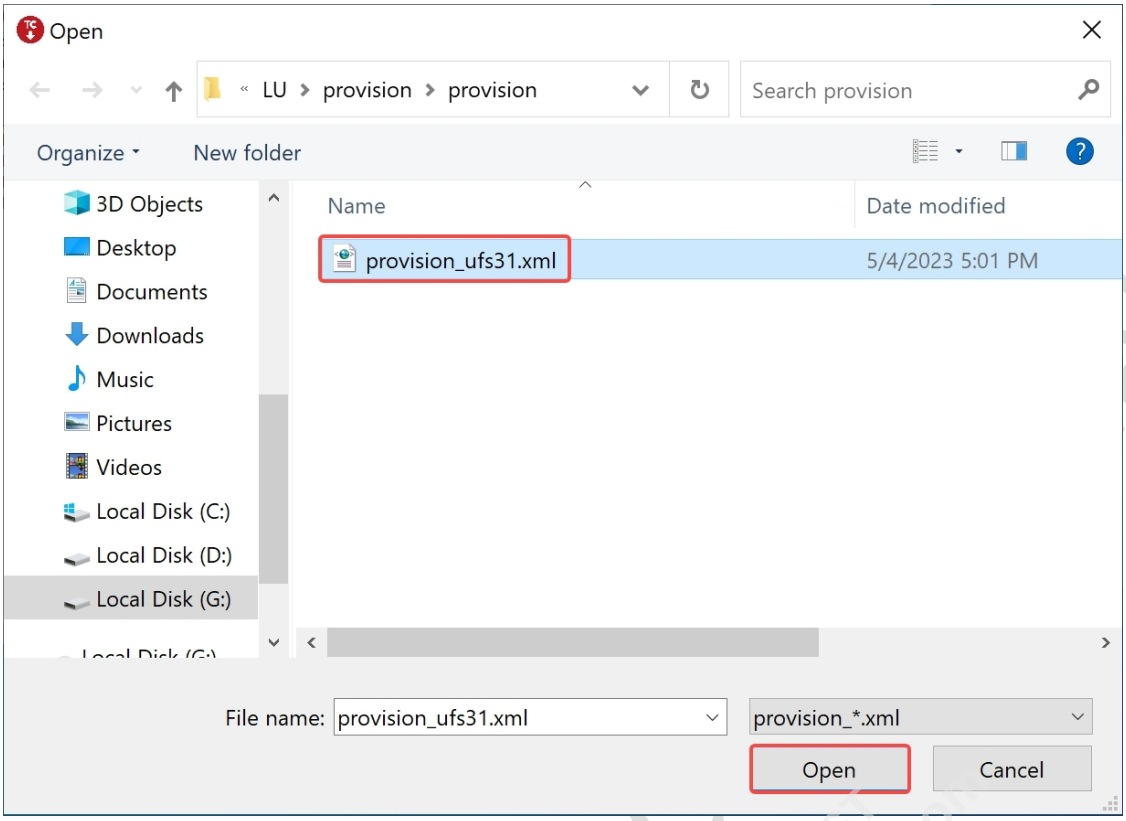


Step 2.Go to the provision directory, select "prog_firehose_ddr.elf" and click on "Open".

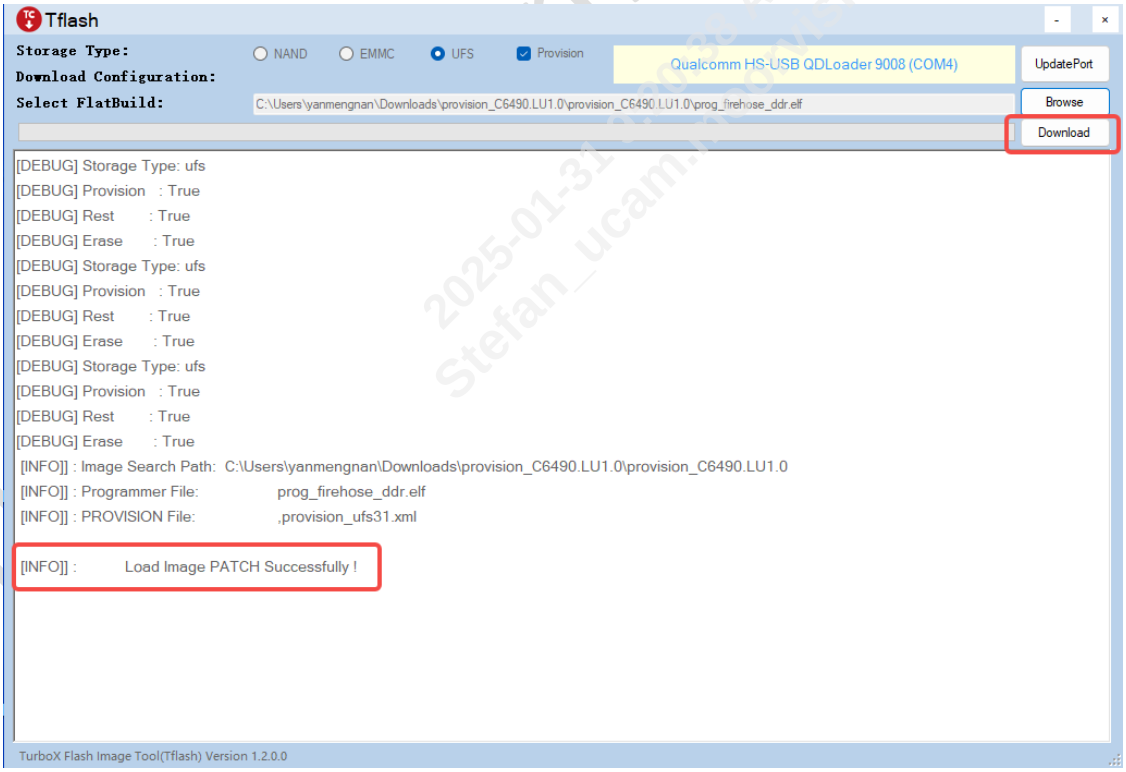
➲ **NOTE:** For specific options, refer to the relevent SDK release note document.



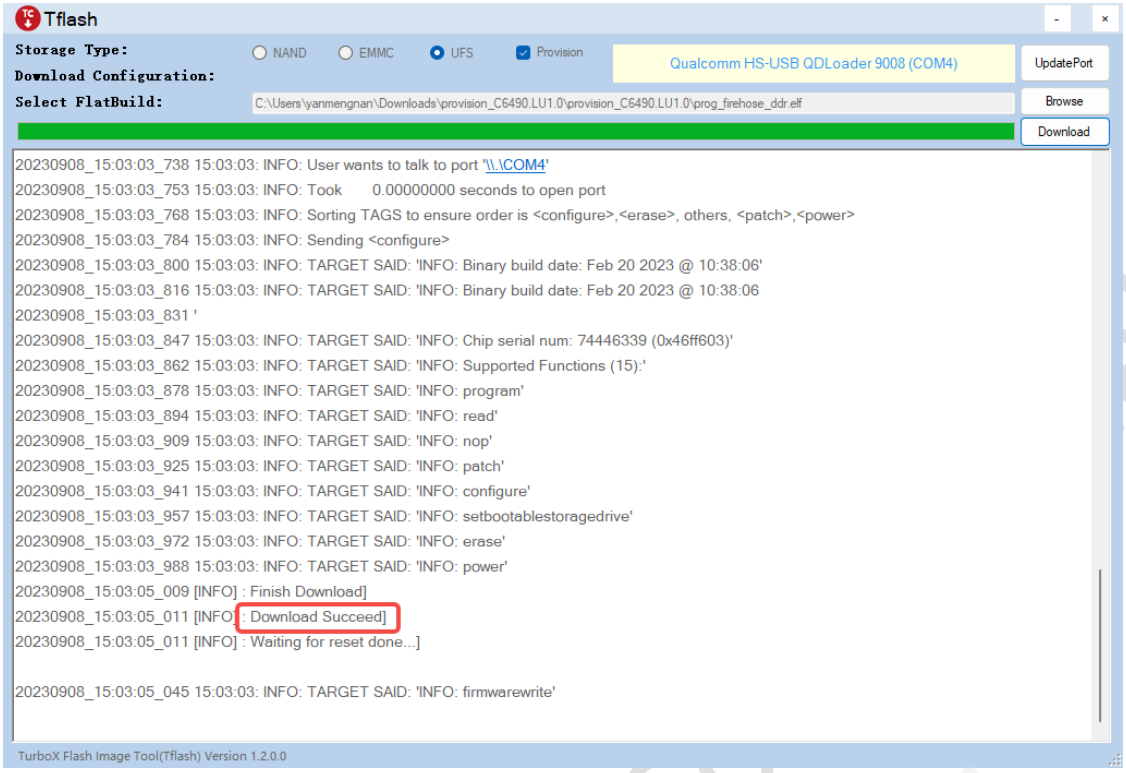
Step 3.Select the "provision_ufs31.xml" file and click on "Open".



Step 4.After the log displays "`[INFO]]: Load Image PATCH Successfully!`", click on "Download" to start the Provision flashing process.



Upon completion of Provision flashing, the interface will display the following log:

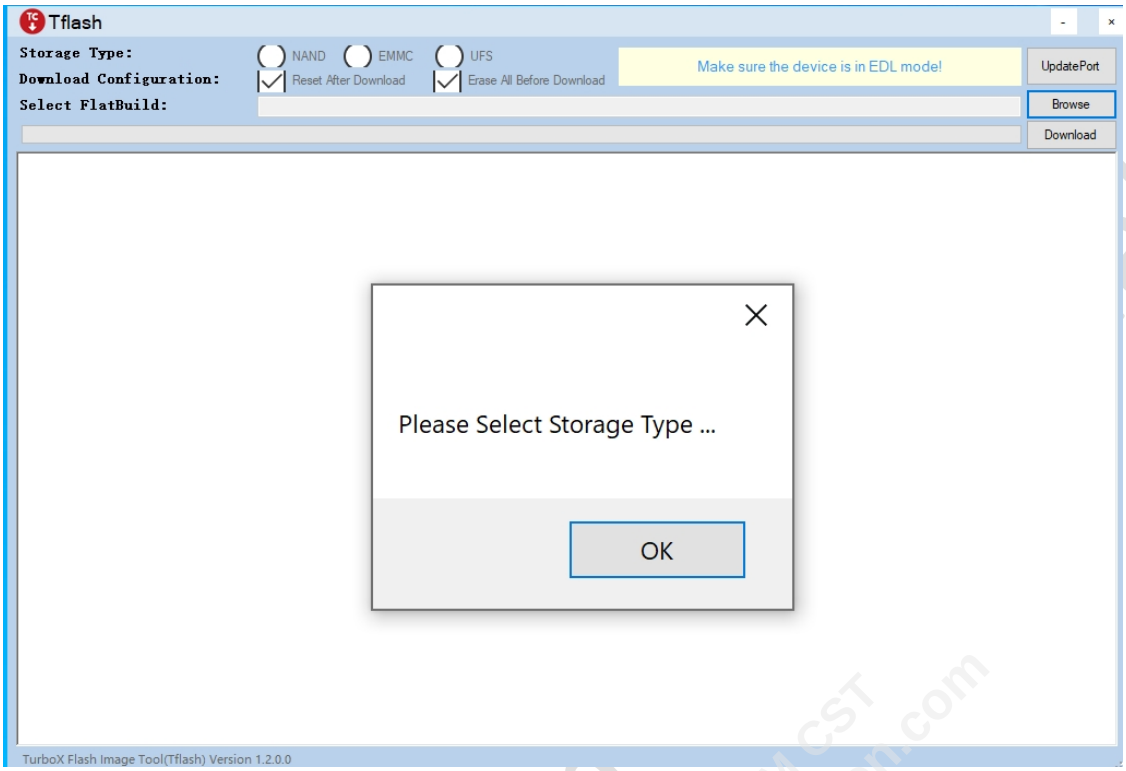


➲ **NOTE:** Upon completion of Provision flashing, you need to manually unplug and plug the power cable and USB cable to restart the device.

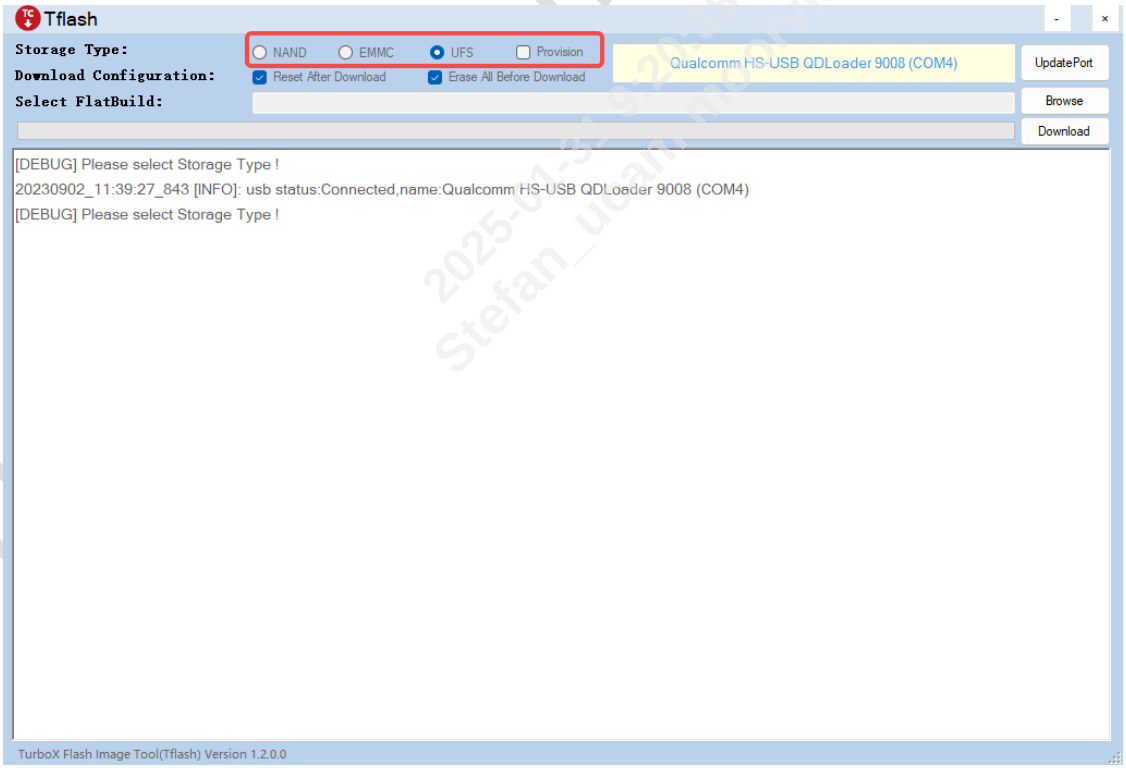
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Chapter 4. Frequently Asked Questions

- **Question 1:** When clicking on "Browse", a "Please Select Storage Type ..." interface pops up.



Solution: Set "Storage Type".

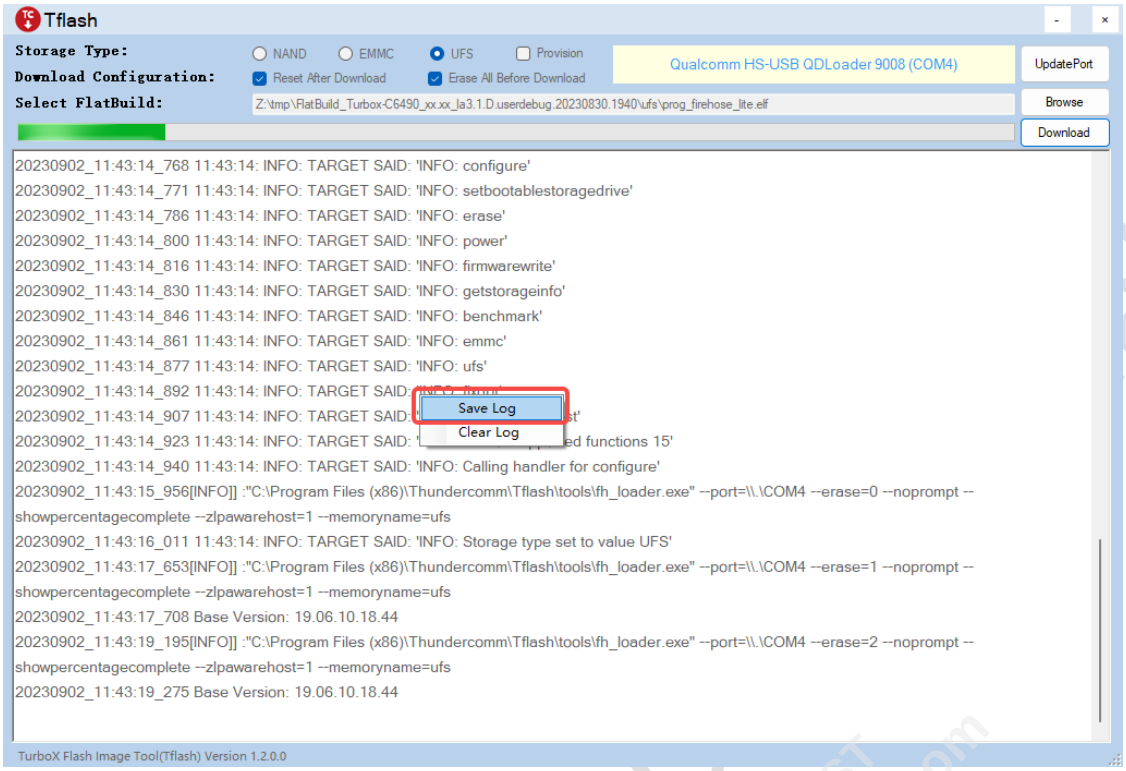


- **Question 2:** After plugging or unplugging the device, the USB Port is not updated in a timely manner.

  Solution: Click on "UpdatePort" to manually update the USB Port.

- **Question** 3: How to save the log from Tflash?

  Solution: In the log output window, right-click on the mouse and select "Save Log".



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